ChatGPT

MCP-Geo Phase 2 evidence-backed benchmark and positioning framework for blocked and hard-to-answer UK public-sector questions dependent on Ordnance Survey data

Executive summary

Evidence-backed findings (facts). Phase 1 established a baseline inventory of **25** currently evidenced, operational UK public-sector question-types that are already being answered with Ordnance Survey data (notably around UPRN/address resolution, planning-policy mapping, routing/network analytics, flood/postcode lookups, and "filter/count/map buildings" use cases). It also surfaced concrete evidence gaps—especially around access-mode detail (API vs download vs enterprise cache), automation of downloads, and areas where operational teams depend on intermediaries. filecite turn0file0

Phase 2 extends that baseline by identifying **blocked, partially answered, or hard-to-answer questions** where the friction is (explicitly) attributed to:
- **manual or time-consuming access and integration** of "raw" geospatial data in real workflows, including planning and housing contexts [1]
- **cross-organisation data format mismatch** that becomes a time cost in emergencies, plus the need to keep Ordnance Survey-derived holdings synchronised across partners [2]
- **address and identifier integration problems** (manual free-text addresses; missing UPRNs; re-keying; post-hoc matching) and the overhead of file-based address data loads versus API-based integration [3]
- **network/routing workflows requiring specialist build steps** that can take days (for example, building a routable network in pgRouting) [4]
- **highways/statistical reporting workflows that still require manual returns or spreadsheets**, and active efforts to replace those manual processes using authoritative road geometry and identifiers [5]
- **licensing constraints** that complicate open publication of "address ↔ UPRN" pairings when both were sourced from licensed Ordnance Survey products (a blocker for open datasets and transparency use-cases). [6]
- **operational data collection and reporting practices still anchored in spreadsheets and manual checking**, even in nationally significant programmes that are UPRN- and AddressBase-dependent. [7]

Structured interpretation (inference). Taken together, the evidence supports a benchmark and positioning framework where MCP-Geo is evaluated and positioned less as "a GIS replacement" and more as a **workflow accelerant** that:
- reduces dependency on scarce specialists for **repeatable, standard queries**, without attempting to replace deeper cartography/data engineering
- bridges the practical gap between **API-first** and **download/cache-first** OS consumption models (both patterns are evidenced as necessary, depending on service context) [8]
- operationalises the government requirement to use UPRN/USRN identifiers by making "identifier-first" location questions easier to ask, answer, verify, and audit. [9]

How Phase 2 builds on Phase 1

Phase 1 baseline (facts). Phase 1 catalogued confirmed, currently-in-use OS-dependent public-sector question patterns and access modes, including: (a) address/UPRN resolution, (b) postcode lookup services with caching patterns, (c) planning mapping use anchored in OS basemaps, (d) routing and network-constraint problems, and (e) building selection/counting tasks. filecite turn0file0

Phase 1 evidence gaps (facts). Phase 1 also documented evidence gaps that are directly relevant to MCP-Geo evaluation design—especially where public descriptions omit whether consumption is via API, bulk resupply/download, vendor-managed supply chain, or enterprise GIS cache, and where automation of downloads wasn't strongly evidenced in named public bodies. filecite turn0file0

Phase 2 extension approach (facts + inference). Phase 2 treats Phase 1 "working use cases" as the **minimum viable reality check**, then looks "one step upstream" at:
- the **unmet question** (what users still can't ask/answer quickly)
- the **workflow friction** (manual steps; specialist bottlenecks; stale copies; licensing constraints; weak discoverability)
- the **stakeholder group** that experiences (or absorbs) the friction
- whether MCP-Geo could plausibly improve the path from question → grounded answer, given the OS data dependency and access model.

Method and evidence rules

Evidence sources and search strategy (facts). This phase prioritised publicly accessible, high-authority material aligned to the user's required source types, specifically: government guidance and standards (including emergency planning and open standards), programme methodology documents, public code repositories used in government delivery, and public-sector geospatial partnership materials. [10]

Rules used to classify each case (facts). Each blocked/hard-to-answer case is categorised as one of:
- **Explicitly stated blocked question** (a source directly states the question is hard/not possible, or states a direct barrier)
- **Explicitly stated workflow pain point** (a source states the workflow is manual/time-consuming/resource-intensive; the implied question is then normalised)
- **Inferred unmet question** (a source documents a cumbersome process; the likely "question the organisation needs answered" is inferred and labelled as such)

Confidence scoring (facts). Confidence scores (0–100) reflect:
- strength of linkage between the question and explicit evidence of difficulty
- strength of linkage to Ordnance Survey data/products (named explicitly vs inferred)
- clarity of stakeholder/team and the nature of the blocker
- reproducibility/verifiability potential for benchmark use.

Coverage and limits

Coverage (facts). Evidence used spans: emergency planning and multi-agency data sharing in England; open standards for UPRN/USRN; planning and housing geospatial landscape; OS documentation for supply and workflow build steps; multiple public-sector delivery and analytical repositories; and public-sector programme documentation relying on AddressBase Premium and UPRN. [11]

Key named organisations represented in the evidence base (facts). Cabinet Office [12], Government Digital Service [13], Central Digital and Data Office [14], Department for Science, Innovation and Technology [15], Building Digital UK [16], Welsh Government [17], Welsh Revenue Authority [18], Valuation Office Agency [19], Ministry of Justice [20], Department for Environment, Food and Rural Affairs [21], GeoPlace [22], Local Government Association [23], Geovation [24], North Yorkshire Council [25], London Borough of Lewisham [26], techUK [27].

Limits (facts).
- Some relevant public procurement attachments and local-government committee document hosts could not be reliably fetched in this environment (server-side restrictions), so Phase 2 weights more heavily towards accessible GOV.UK, OS documentation, and public repositories.
- Many public write-ups (including some case studies) describe outcomes and tooling but omit precise OS access pattern details; where access pattern is inferred, it is labelled as inferred rather than asserted. [28]

Questions public bodies appear to want to answer, but cannot yet answer well because OS data access or usability is a barrier

Patterns in blocked or partially answered OS-dependent questions (facts). Across the evidence base, repeated "hard-to-answer" patterns cluster into a small number of question families:

Address and identifier integration as a prerequisite for almost everything else. Government open standards require that systems storing/publishing property and street information use UPRN/USRN identifiers, and link them to definitive information in Ordnance Survey products (under public sector entitlement). [29] Yet emergency guidance warns that managing multiple address forms incorrectly leads to duplication and real effort/resourcing impacts, and that Ordnance Survey product updates must be maintained to avoid version conflicts across users. [2] The GeoPlace integration checklist further makes explicit that free-text/manual addresses are unverified and duplicative, and that file-based address data import is "complex", with non-trivial overhead that APIs can reduce. [30]

Cross-organisation crisis questions are slowed by format and version mismatch. Cabinet Office emergency guidance explicitly states that data presentation and incompatible formats can become a barrier; it can be time-consuming to convert multi-source data into usable data, and software/compatibility issues can delay information sharing. [2] Inference: many "urgent" location questions (who is affected, where are they, what support is needed) are answerable in principle but remain slow in practice because address/UPRN joins and boundary overlays cannot be performed quickly by non-specialists.

OS supply and "local copy" handling creates data management friction that shows up as operational delay. AddressBase supply is divided into non-geographic chunks (where nearby features can be in different volumes, and features from different geographies in the same volume), with update mechanisms (COU) that require careful replacement logic—especially for geographic chunked updates where tile replacement semantics apply. [31] Inference: these mechanics create predictable failure modes: stale copies, partially updated holdings, and workflow gatekeeping by those who understand the supply mechanics.

Network/routing questions are "answerable" but can be operationally blocked by build steps. OS routing guidance explicitly states that loading data into pgRouting and building a network can be time consuming and can take 2–3 days, with a recommendation to build routines that run overnight. [4]

Inference: that timescale converts many desirable "operational" routing questions (school transport, inspections, waste, welfare checks) into specialist, scheduled exercises rather than interactive questions.

Planning/housing workflows show both data interoperability problems and capacity constraints. The planning and housing landscape review explicitly states that access to raw geospatial data in an efficient manner is often manual and time-consuming, and highlights software and system interoperability constraints that "lock" users into tools that can't communicate. [1] It also reports that a majority of local authorities interviewed cited lack of geospatial skills/resources as a top barrier, with recruitment/retention challenges compounding this. [1]

Highways/statutory reporting and reclassification still contains manual returns and spreadsheets. GeoPlace's write-up of highways-related work describes piloting the use of detailed road geometry to identify maintainable segments **without requiring manual submissions**, and explicitly acknowledges the pain of reclassification spreadsheets. [5]

Programme reporting still depends on spreadsheets and manual checking even when OS identifiers are central. BDUK's methodology documents explicitly state that supplier delivery management information is submitted as spreadsheets and that some intervention data is updated via delivery tracker spreadsheets "manually checked by analysts"; they also note UPRN-level data gaps that prevent exact reporting in some cases, requiring estimation assumptions. [32] Separately, BDUK's "premises in plans" user guide states the release is optimised for technical users (programming), may not suit Excel, and that missing UPRNs can be due to new builds not included in the AddressBase Premium epoch used. [33]

Where current OS access models create friction (facts). The evidence points to three recurring friction points that show up differently by context:

- **API-first (transactional) access** is repeatedly positioned as a way to avoid file-handling overhead (GeoPlace explicitly describes APIs removing overhead of obtaining files, loading, and validation). [30]
- **Download/cache-first (batch/enterprise) access** is structurally complex for some OS products (chunking, COU semantics, periodic updates). [34]
- **Hybrid access patterns** arise where organisations need authoritative identifiers and addresses, but also need repeatable analytics on stable copies (for example, address matching pipelines that download and transform AddressBase Premium for matching, including version management and chunking for resource constraints). [35]

Where specialist GIS bottlenecks appear most severe (facts). Planning/housing evidence explicitly links capability gaps and interoperability constraints to barriers in local authority contexts. [1] Local-government practice evidence shows GIS teams acting as intermediaries even for statutory or recurrent processes (for example, a county council process describes housing development data as having no single source and being checked/adjusted through GIS workflows and audit spreadsheets, including cross-checks against OS basemaps and LLPG signals). [36]

Questions especially suitable for a conversational OS-backed interface (inference grounded in evidence). The strongest candidates share four traits supported by the evidence base:
- they begin with a **natural-language location reference** (postcode, address, place name) that must be resolved to UPRN/geometry [37]
- they require a **repeatable, templated spatial operation** (point-in-polygon; within distance; intersects; nearest; count-by-area), rather than bespoke cartography [38]
- they are slowed by **manual integration steps** (spreadsheets, re-keying, mismatch, manual returns),

not by the absence of any data at all [39]
- they need an **audit trail** and clear versioning/lineage (explicitly raised in emergency guidance and address ecosystem transformation narratives). [40]

Mandatory output Table 1 — Blocked or unmet questions (with evidence, workarounds, blockers, and OS dependency).

blocked_question_id	organisation_name	organisation_type	department_name	team_name	service_area
BQ01	Local Resilience Forum partners (generic)	multi-agency partnership	N/A	N/A	emergency planning & response
BQ02	Local Resilience Forum partners (generic)	multi-agency partnership	N/A	N/A	emergency data linkage
BQ03	Public-sector organisations maintaining OS-derived holdings (generic)	public sector (generic)	N/A	N/A	data management & integrity
BQ04	UK public bodies publishing/storing property/street data (generic)	public sector (generic)	N/A	N/A	data standards compliance
BQ05	Local authority service teams (generic)	local authority	N/A	N/A	citizen-facing forms & service requests

blocked_question_id	organisation_name	organisation_type	department_name	team_name	service_area
BQ06	Local authority address custodians & service owners (generic)	local authority	N/A	Authority Address Custodian	data quality remediation
BQ07	Local authority digital/data teams (generic)	local authority	N/A	IT / system admin	data integration
BQ08	Public bodies consuming AddressBase (generic)	public sector (generic)	N/A	N/A	data engineering
BQ09	Public bodies using geographic chunk COU (generic)	public sector (generic)	N/A	N/A	data updates
BQ10	Defra digital services (multiple)	central government	Defra	N/A	address lookup services
BQ11	Ministry of Justice analytical services (generic)	central government	MoJ	data/ analytics engineering	address matching at scale

blocked_question_id	organisation_name	organisation_type	department_name	team_name	service_area
BQ12	Public-sector developers using OS APIs (generic)	public sector (generic)	N/A	N/A	infrastructure enablement
BQ13	Local authorities using routing analytics (generic)	local authority	N/A	GIS / data teams	routing optimisation
BQ14	Highways authorities (generic)	local authority	N/A	streetworks/ highways	road maintainabilit & stats
BQ15	Highways authorities (generic)	local authority	N/A	streetworks/ highways	reclassificatio
BQ16	Local planning authorities (generic)	local authority	planning	N/A	planning & housing
BQ17	Local planning authorities (generic)	local authority	planning	N/A	planning systems
BQ18	Local planning authorities (generic)	local authority	planning	N/A	skills/capacity

blocked_question_id	organisation_name	organisation_type	department_name	team_name	service_area
BQ19	London Borough of Lewisham	local authority	multiple	multiple	waste, SNN, assets, land
BQ20	London Borough of Lewisham	local authority	regeneration/ estates	N/A	asset & land management
BQ21	London Borough of Lewisham	local authority	children's services / public health	Support for Troubled Families team	multi-dataset linkage
BQ22	Valuation Office Agency	central government agency	data architecture	N/A	council tax & NDR address data
BQ23	Building Digital UK	central government directorate	analysis & evaluation	statistics production	broadband delivery statistics
BQ24	Building Digital UK	central government directorate	analysis & evaluation	statistics production	comparability/ coverage
BQ25	Building Digital UK	central government directorate	programme transparency	N/A	public/local council queries

blocked_question_id	organisation_name	organisation_type	department_name	team_name	service_area
BQ26	Building Digital UK	central government directorate	programme transparency	N/A	data completeness
BQ27	Risk Management Authorities in Wales (generic)	devolved/local public bodies	N/A	N/A	flood & coasta risk appraisal
BQ28	North Yorkshire Council	local authority	data & intelligence / planning	GIS Team	elections & housing forecasting
BQ29	Welsh Revenue Authority	devolved public body	service design/ data	Data lead for Service Design	land/property data platform
BQ30	Public-sector open data publishers (generic)	public sector (generic)	N/A	N/A	open data & transparency

How MCP-Geo should be positioned for different stakeholder groups

Stakeholder mapping (facts + inference). Government open standards explicitly call out likely user groups for property/street identifiers (publishers of government data, data scientists, analysts, developers). [29] Planning/housing evidence shows many local authorities facing skills gaps and tool interoperability constraints. [1] Emergency guidance highlights multi-agency conversion and versioning burdens. [2] These sources, combined with operational code and pipeline evidence (Defra and MoJ), support a differentiated stakeholder map focused on **who asks**, **who answers**, and **who maintains** OS-dependent spatial capability. [47]

Differentiated value propositions (inference grounded in evidence). MCP-Geo positioning should vary by stakeholder "job to be done", because the evidence shows friction is not uniform:

- For **operational users and service teams**: "validated address/UPRN answers fast; fewer errors and duplicates" (strongly grounded in GeoPlace checklist and emergency guidance). [48]
- For **GIS teams**: "reduce the queue by handling routine, well-scoped questions with auditability; keep specialists on complex work" (grounded in documented skills/capacity constraints and GIS gatekeeping patterns). [49]
- For **data/platform teams**: "bridge API/download complexity; improve pipeline repeatability; make lineage/versioning explicit" (grounded in AddressBase supply mechanics and public pipelines). [50]
- For **emergency planning/resilience**: "join-ready datasets and rapid affected-premises answers; fewer format delays" (grounded in Cabinet Office guidance). [2]
- For **highways/statutory reporting**: "reduce spreadsheet/manual returns; data-quality feedback loops" (grounded in GeoPlace streets evidence). [5]
- For **programme/statistics teams**: "reduce spreadsheet bottlenecks; surface data gaps/ unknowns; improve reproducibility" (grounded in BDUK methodology). [7]

Mandatory output Table 2 — MCP-Geo stakeholder map (positioning by stakeholder type).

stakeholder_id	stakeholder_group	organisation_archetypes	department_or_team_archetypes	typical_questio
SG01	GIS and geospatial teams	local authorities; agencies	GIS; spatial analytics; BI	"Which premise is within this bo changed since
SG02	Digital/data platform teams	departments; councils; shared services	data engineering; cloud ops; API gateways	"How do we ing dataset/version expose it safely
SG03	Service analysts and statisticians	national programmes; councils	analysis & evaluation; performance	"How many pre eligible?" "Wher
SG04	Emergency planning/ resilience teams	LRFs; local authorities; responders	resilience; emergency planning	"Who is in the a vulnerable sites

stakeholder_id	stakeholder_group	organisation_archetypes	department_or_team_archetypes	typical_questio
SG05	Planning/housing specialists	LPAs; combined authorities	planning policy; development management	“Which constra “What is the evi
SG06	Highways and streetworks teams	highways authorities	street gazetteer custodians; asset mgmt	“What is mainta count?”
SG07	Environmental/ flood risk teams	devolved/local RMAs; agencies	FCERM appraisal; adaptation	“How many pro the classificatio
SG08	Operational delivery teams	councils; agencies	customer services; field ops	“What’s the cor it?”
SG09	Estates/asset managers	councils; public estates	regeneration; property; FM	“What do we ov site?”
SG10	SROs/AI/ innovation leads	departments; councils	transformation; strategy	“Where do we g are low-risk/hig

Recommended first benchmark pack for MCP-Geo evaluation

Benchmark design principles (facts + inference). Evidence indicates that the highest-value benchmarks should:
- start from **UPRN/USRN standards compliance and operational use**, because government mandates identifiers and they underpin multi-dataset linkage [9]
- include both **API-first** and **download/cache-first** patterns, because both are evidenced as necessary, and each has distinct friction modes (file overhead vs chunk/update semantics vs deployment burden) [52]
- test **reproducible spatial operations** (counts, joins, overlays) with explicit handling of “unknowns” and update cycles, mirroring how public reporting currently deals with data gaps and epoch lag [7]
- include at least one “hard” benchmark that reflects **specialist time costs**, such as routing network build and reuse, because evidence explicitly flags multi-day build cycles. [4]

Recommended benchmark suite (structure). This report provides a **75-question test bank** (Table 3).
- Priority ranks **1–25** = **top 25 must-test questions** (high cross-sector value + strong OS dependency + high benchmark verifiability)
- Priority ranks **26–75** = next 50 strong candidates (valuable but either more context-dependent, higher governance burden, or harder to verify at scale).

Recommended first benchmark pack for MCP-Geo evaluation (inference grounded in evidence). If a first-run evaluation pack must be smaller than 25, the evidence suggests starting with **12 benchmarks** that cover the most recurrent OS-dependent friction types:
- address lookup and batch UPRN matching (GeoPlace checklist; Defra code duplication) [53]
- multi-agency "affected premises" joins (Cabinet Office guidance) [2]
- routing query using pre-built network + explanation of build cost/constraints (OS routing methodology) [4]
- BDUK "premises status" Q&A and "missing because epoch" explanations (BDUK user guide/QMI) [54]
- highways maintainability anomaly checks (GeoPlace streets evidence) [5]
- property-platform identifier alignment and "unmatched" quantification (Welsh Revenue Authority statement) [46]

Mandatory output Table 3 — MCP-Geo benchmark/test bank (top 25 must-test + next 50 candidates).

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T001	"What is the UPRN, coordinates, and authoritative formatted address for: 'Flat 2, 10 High Street, [postcode]'?"	BQ04,BQ05,BQ10	Operational delivery	cross-sect
T002	"Batch match these 5,000 free-text addresses to UPRNs; produce match rate, unmatched list, and top failure reasons."	BQ06,BQ11	Data platform/GIS	cross-sect
T003	"Which records in this dataset have no UPRN, and what is the recommended remediation route?"	BQ06,BQ04	Service analysts	cross-sect
T004	"Given a UPRN, return whether it is residential/ commercial/other using authoritative classifications."	BQ04,BQ26	Analysts	infrastruct programm

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T005	“Draw a 500m buffer around this incident point and list affected UPRNs; join to ‘vulnerable households’ dataset and summarise counts.”	BQ01,BQ02	Emergency planning	resilience
T006	“Explain how address duplication occurs (Local Property Identifier vs Delivery Point Address) and propose a dedupe approach using UPRN.”	BQ02	Emergency/data governance	resilience
T007	“Are our OS-derived holdings out of date? Given last refresh dates, identify which datasets are stale and what to update next.”	BQ03,BQ09	Data platform	cross-sect
T008	“Given AddressBase geographic chunk COU for tiles X,Y, show the safe update steps (delete/replace) and the QA checks.”	BQ09	Data engineering	cross-sect
T009	“Which OS product(s) should we use to validate UPRNs and USRNs for this service, and why?”	BQ04	Digital/data teams	cross-sect
T010	“Standardise address lookup behaviour across three services: define consistent request/ response schema and error handling.”	BQ10	Developers	environme governme services

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T011	“Transform AddressBase Premium download into parquet suitable for matching; confirm version_id in config and note update triggers.”	BQ11	Data platform	justice/ analytics
T012	“Deploy a minimal OS-backed API wrapper with credential handling and audit logging; list required inputs and controls.”	BQ12	Platform teams	cross-sect
T013	“Given two UPRNs, compute shortest road-network route and travel distance, respecting restrictions.”	BQ13	GIS/ops	local servi
T014	“Explain why building the routable network is slow and propose an overnight build schedule and reuse strategy.”	BQ13	GIS/platform	local servi
T015	“Which road segments in this area are publicly maintainable and what is the total length by class?”	BQ14	Highways	transport
T016	“Given anomaly list between NSG and OS Highways, prioritise fixes and explain likely causes.”	BQ14,BQ15	Highways custodians	transport

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T017	“For this planning site polygon, list intersecting constraints and produce an evidence summary.”	BQ16,BQ17	Planning	planning/ housing
T018	“Why is this planning data question currently slow? Identify which steps are manual and propose automation.”	BQ16,BQ17,BQ18	Planning/digital	planning/ housing
T019	“From a council asset spreadsheet, link each row to a UPRN and map assets within 200m of a flood extent.”	BQ20,BQ27	Estates/ environment	local gov
T020	“For flood appraisal: count properties at risk and generate a targeted verification list (site walkover vs desktop checks).”	BQ27	Flood risk	environme
T021	“For a given UPRN, return whether it is included in BDUK premises base and explain missing/ new-build logic.”	BQ25,BQ26	Analysts/councils	broadband
T022	“Summarise delivered premises by local authority; identify ‘unknown geography’ records and suggest match strategy.”	BQ24	Analysts	broadband
T023	“Given housing development datasets from multiple systems, normalise and spatially summarise by polling district.”	BQ28	GIS/planning	local gov

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T024	“Build a coherent property platform prototype: link council tax-like records + land registry-like polygons via UPRN.”	BQ29,BQ22	Data/policy	taxation/ property
T025	“Can we publish this dataset openly with UPRNs? Assess licensing lineage and propose compliant publication options.”	BQ30	Data governance	cross-secto
T026	“Given 10 messy address strings, produce canonical forms and explain parsing choices.”	BQ05,BQ11	Operational/ analysts	cross-secto
T027	“Detect duplicates in a case-management dataset caused by manual address entry and propose merges via UPRN.”	BQ05,BQ02	Ops/data	local gov
T028	“Validate a list of USRNs and identify which are invalid/out of date.”	BQ04	Highways/data	transport
T029	“Given a citizen report location (lat/ long), return nearest address, UPRN, and council service area.”	BQ01,BQ05	Customer service	local gov
T030	“Show the minimal dataset set required to answer ‘within 1km of school’ questions reliably.”	BQ16,BQ04	Policy/analysts	cross-secto
T031	“Given an address list, assign local authority, ward, and small area geographies by point-in-polygon.”	BQ01,BQ03	Analysts	cross-secto

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T032	“Identify premises that exist in OS address data but are missing from an internal service dataset; quantify impact.”	BQ03,BQ06	Data governance	cross-sect
T033	“Given a dataset refresh schedule, produce a monitoring plan and alert rules for missed updates.”	BQ03,BQ09	Platform	cross-sect
T034	“Find the authoritative dataset fields needed to distinguish ‘demolished/derelict/ new build’ premises.”	BQ26	Analysts	infrastruct
T035	“For a given area, list ‘provisional’ addresses and estimate occupancy status.”	BQ28	Planning/elections	local gov
T036	“Compute recycling route zones and show which households fall into each schedule.”	BQ19	Service ops	local gov
T037	“Generate a ‘data quality heatmap’ of address matching failures by area to target cleansing.”	BQ06,BQ11	Data quality	cross-sect
T038	“For a service request dataset, show how many records are non-joinable due to missing UPRN and why.”	BQ06	Service analysts	local gov

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T039	“Explain how UPRNs support obfuscation when sharing between organisations and what fields to exclude.”	BQ01	Emergency/info gov	resilience
T040	“Given a highway segment, explain why it is flagged as not publicly maintained using geometry context.”	BQ14	Highways	transport
T041	“Generate a candidate list of roads needing custodian review before March snapshot deadline.”	BQ14	Highways	transport
T042	“Summarise planning datasets currently not FAIR and propose minimum metadata fields to fix.”	BQ16,BQ17	Data policy	planning/ housing
T043	“For a proposed development, list nearby utilities/ amenities datasets needed and warn about licensing.”	BQ16,BQ30	Planning/infra	planning/ housing
T044	“Audit a dataset for UPRN/USRN field presence and generate a compliance scorecard.”	BQ04	Data governance	cross-sect
T045	“Given 100 UPRNs, confirm which are genuine using a trusted source and report any invalids.”	BQ04	Developers/ analysts	cross-sect

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T046	“For an incident area, list all care settings and schools (premises types) and export for responders.”	BQ01	Emergency	resilience
T047	“Identify candidate properties for targeted comms by filtering on classification (e.g., HMOs/blocks).”	BQ04,BQ26	Service ops	cross-sect
T048	“Given a data request, generate a minimum viable data sharing agreement header/ schema.”	BQ01	Emergency/digital	resilience
T049	“Provide a ‘citizen enquiry’ answer: why might we not respond to a premises query and what to include.”	BQ25	Public comms	broadban
T050	“Crosswalk a bespoke property ID list to UPRNs using fuzzy + rules; compare match to baseline.”	BQ11,BQ06	Data engineers	cross-sect
T051	“Estimate uncertainty: where UPRN-level gaps exist, produce ‘known/unknown’ split and impact statement.”	BQ24	Analysts	broadban
T052	“Detect when a system’s address load silently failed (no feedback) and list consequences.”	BQ07	Platform/ops	cross-sect

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T053	“Recommend whether to use OS Places API or bulk AddressBase for ‘out of area’ scenarios.”	BQ07	Ops/digital	cross-sect
T054	“Create a map-ready GeoJSON export of matched UPRNs with minimal sensitive fields.”	BQ01,BQ11	Emergency/IG	cross-sect
T055	“Generate an audit log narrative: which OS datasets/versions were used for this answer?”	BQ03,BQ11	Governance	cross-sect
T056	“Given a boundary change (old vs new geographies), propose method to compare time series consistently.”	BQ28	Analysts	local gov
T057	“Explain ‘tile vs non-geographic chunk’ implications for local authority area-of-interest extracts.”	BQ08	Data engineers	cross-sect
T058	“Create a ‘data drift’ detector: compare two AddressBase epochs and highlight high-change areas.”	BQ03,BQ26	Analysts	local gov
T059	“For an address, retrieve parent UPRN and explain hierarchy (e.g., flats in a block).”	BQ22	Analysts	taxation/ property
T060	“Given a complex site (hospital campus), show why one-to-one mapping fails and propose junction table design.”	BQ22	Data architects	health/ property

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T061	“Identify which datasets could legally be made open under OGL vs which require PSGA entitlements.”	BQ30,BQ04	Data governance	cross-sect
T062	“Provide a ‘minimum viable OS-backed answering service’ architecture for a local authority.”	BQ12,BQ07	Transformation/ platform	local gov
T063	“Given service constraints, propose when local caching is operationally necessary vs API sufficient.”	BQ07,BQ03	Platform/GIS	cross-sect
T064	“For a council tax file, standardise addresses and flag which cannot be reconciled to authoritative sources.”	BQ22	Tax/property	local gov
T065	“Summarise the likely human time saved if manual R199B-like processes are avoided for this authority.”	BQ15	Highways leadership	transport
T066	“Generate a ‘what changed’ briefing for elected members based on housing development geospatial changes.”	BQ28	Leadership	local gov
T067	“Create a reproducible notebook-style query plan (SQL) equivalent to the conversational answer.”	BQ11,BQ24	Analysts	cross-sect

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T068	“Explain when ‘manual site walkover’ is still necessary and how to optimise it using OS-backed evidence.”	BQ27	Flood risk	environme
T069	“Given a sensitive dataset, show a de-identified join strategy using UPRN without leaking personal data.”	BQ01,BQ02	IG/data	cross-sect
T070	“Identify candidate data quality KPIs for address matching pipelines (match rate, drift, latency).”	BQ11,BQ03	Platform	cross-sect
T071	“For the same query, show differences between live API answers and cached holdings; explain why.”	BQ03,BQ07	Platform/GIS	cross-sect
T072	“Produce ‘data lineage’ statement: LLPG → OS Places → internal system; identify transformation risks.”	BQ22,BQ03	Data architects	taxation/ property
T073	“Create a training-style ‘geospatial for data scientists’ primer using an OS-dependent example.”	SG02,SG03	Skills	cross-sect
T074	“Detect ‘legacy address truncation’ risk and propose field schema changes aligned to standards.”	BQ22	Data architecture	taxation/ property

test_id	benchmark_question	based_on_real_use_case_or_blocked_question_ids	stakeholder_group	sector
T075	“Recommend a phased adoption plan: start with UPRN lookups, then joins, then routing.”	all	Transformation	cross-sect

Risks, caveats, and evidence gaps

Evidence gaps that remain material (facts).
- **Access-mode detail is still often under-specified in public narratives**: many descriptions of outcomes do not confirm whether OS data was consumed via OS APIs, downloads/resupply, enterprise GIS cache, or vendor supply chain; Phase 1 explicitly recorded this as a core evidence gap. filecite turn0file0
- **Licensing constraints can be a hard blocker in open publication contexts**, particularly where a dataset’s “address ↔ UPRN” pairing provenance traces to licensed OS products; this risk is explicitly described by respondents discussing the difficulty of adding UPRNs to open datasets under certain licensing conditions. [6]
- **Some high-value workflows are inseparable from information governance and privacy constraints**, especially where UPRN enables household-level linkage (emergency vulnerability, safeguarding, multi-agency support). Emergency guidance explicitly frames interoperability and UPRN use alongside data sharing agreements and format governance. [2]
- **Routing/network analytics carry unavoidable build-time and compute costs** when organisations are responsible for building their own routing networks, with multi-day build steps explicitly stated in OS guidance. [4]
- **Planning/housing evidence strongly signals skills gaps and interoperability constraints**, but does not always identify the precise OS dataset involved in each constrained decision workflow; OS dependence is sometimes foundational but implicit. [1]

Risks for MCP-Geo evaluation design (inference grounded in evidence).
- If benchmarks rely only on “easy” API lookups, evaluation will miss the hardest and most operationally costly friction documented in the evidence: chunk/update semantics, pipeline build steps, and spreadsheet-based reporting bottlenecks. [55]
- If benchmarks ignore governance and privacy, evaluation will not represent the real adoption constraints in emergency/resilience and cross-service household linkage contexts, where UPRN is powerful precisely because it enables linkage. [56]
- If MCP-Geo is positioned as a broad “answer anything” geospatial assistant, stakeholders may expect it to adjudicate planning judgement or replace statutory processes; evidence suggests the strongest near-term positioning is **repeatable routine answers + auditability + guided workflows**, not replacing professional judgement. [57]

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